

A Hybrid TF-IDF–SBERT Framework with Rule-Based Skill Verification for Explainable, Bias-Mitigated Resume Screening

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Abstract— Automated Applicant Tracking Systems (ATS) commonly rely on either rigid keyword matching or opaque deep-learning embeddings, and both approaches present distinct shortcomings for high-stakes hiring decisions. Keyword-based systems suffer from vocabulary mismatch, while transformer-based semantic models can overestimate candidate suitability when contextual similarity is not supported by genuine technical competence. This paper proposes a hybrid recruitment framework that fuses sparse lexical representations (TF-IDF), dense contextual embeddings (SBERT, all-MiniLM-L6-v2), and a deterministic rule-based skill-verification layer to balance semantic flexibility with explicit competency validation. An identity-fair preprocessing stage built on spaCy Named Entity Recognition anonymizes sensitive demographic attributes prior to feature extraction, and a safety-gate mechanism penalizes candidates whose required-skill match ratio falls below a fixed threshold. Weighted score fusion (45% SBERT, 40% skill matrix, 15% TF-IDF) followed by sigmoid calibration classifies candidates into four categories: Strong Match, Partial Match, Low Match, and Failed Validation. The framework was evaluated on 120 Information Technology and software-engineering resumes drawn from a public Kaggle dataset, ranked against five constructed job descriptions and benchmarked on a stratified, manually annotated sample of 75 resume–job description pairs. While a semantic-only SBERT baseline achieved the highest Mean NDCG (0.9689), the proposed hybrid framework improved Recall to 0.6167 over the TF-IDF baseline (0.5500) while retaining a high Precision of 0.9737, indicating a more practical balance between candidate discovery and technical-qualification enforcement for real-world recruitment settings.

Keywords—resume screening; Applicant Tracking Systems; Sentence-BERT; TF-IDF; explainable AI; algorithmic bias mitigation; hybrid ranking; NDCG.

I. INTRODUCTION

The growth of digital job applications has far outpaced the capacity of human resource departments to review them manually. To cope with this volume, organizations increasingly rely on automated Applicant Tracking Systems (ATS) to parse, sort, and rank candidate resumes against job descriptions. However, as Al-Otaibi and Ykhlef [1] observed, traditional rule-based screening tools depend heavily on keyword indexing and static regular-expression filters. This rigid approach causes vocabulary mismatch, in which strong candidates who describe their experience using different terminology are misclassified and dropped from consideration.

Modern deep-learning methods address this language-fragmentation problem by mapping text into dense vector spaces that capture overall meaning rather than exact wording. While transformer-based encoders handle lexical variation well, relying on sentence embeddings alone introduces a different risk, which we term prose masking: a resume written in sophisticated language may receive an undeservedly high similarity score for a technical role even when the candidate lacks the required certifications, experience, or practical skills. Deep transformer networks also act as non-linear black boxes that offer no direct explanation for their rankings, which makes it difficult to detect bias or satisfy compliance audits in high-stakes hiring.

To address these shortcomings, this paper presents a hybrid recruitment framework for automated candidate screening that combines sparse lexical text mining, dense transformer

representations, and explicit rule-based verification. The architecture forms a multi-tiered ranking pipeline that connects syntax and semantics: a sublinear log-scaled TF-IDF baseline [2], pre-trained SBERT contextual embeddings [3] generated with the all-MiniLM-L6-v2 bi-encoder, and a deterministic symbolic skill matrix. The SBERT channel captures broader semantic meaning, the TF-IDF channel preserves lexical alignment, and the symbolic skill matrix directly validates domain-specific technical competencies against curated prerequisite dictionaries.

To mitigate irrelevant textual noise and keyword-stuffing behavior, the framework incorporates a rule-based safety gate that acts as a programmatic competency threshold. If a candidate's required-skill match ratio falls below 20%, a fixed penalty of -0.25 is applied to push non-compliant applications toward the bottom of the ranking. The adjusted scores are then passed through a logistic sigmoid transformation and grouped into four classification categories, using a calibrated 65% score cutoff: Strong Match, Partial Match, Low Match, and Failed Validation. Ranking quality is evaluated with standard information-retrieval metrics, including Normalized Discounted Cumulative Gain (NDCG) [5].

To promote fairness and reduce historical bias, the framework applies an identity-fair preprocessing layer built on the spaCy parser [6], which removes sensitive demographic information before feature extraction. It additionally supports transparency and recruiter-side inspection through optional explanation and auditing mechanisms [7], [8], generating audit logs that surface influential textual features, missing technical skills, and the key factors behind each candidate's ranking. Collectively, these

components aim to preserve high screening accuracy while improving the accountability and trustworthiness of automated hiring decisions.

II. LITERATURE REVIEW

Automated Applicant Tracking Systems have evolved from keyword-based retrieval methods toward semantic matching approaches driven by deep learning. Early studies on automated resume screening documented several limitations of rule-based recruitment systems. Al-Otaibi and Ykhlef [1] observed that keyword-dependent screening treats words as independent features and fails to recognize semantic equivalence between related terms, causing qualified candidates to be overlooked when their resumes use terminology that differs from the job description.

Many early ATS platforms trace their technical foundation to the Vector Space Model (VSM) proposed by Salton et al. [2], in which documents are represented as sparse vectors weighted by Term Frequency–Inverse Document Frequency (TF-IDF) and ranked via cosine similarity against a job description. Although computationally efficient and widely adopted, such keyword-based systems remain vulnerable to vocabulary mismatch and keyword stuffing, in which applicants artificially inflate their ranking by inserting job-related terms without demonstrating genuine expertise [1], [2].

To overcome the limits of sparse lexical representations, researchers have increasingly adopted dense semantic embeddings. Devlin et al. [4] introduced Bidirectional Encoder Representations from Transformers (BERT), which generates context-aware representations by processing text bidirectionally and captures semantic relationships that bag-of-words models miss. However, standard BERT cross-encoder architectures require joint processing of each job description and resume pair, which makes large-scale screening computationally expensive.

To improve scalability, Reimers and Gurevych [3] proposed Sentence-BERT (SBERT), a Siamese network architecture that produces fixed-dimensional sentence embeddings while preserving semantic similarity. Because resumes and job descriptions are encoded independently, SBERT enables efficient cosine-similarity comparisons while retaining strong semantic understanding. Building on this idea, Lavi et al. [9] introduced conSultantBERT, a domain-specific adaptation of SBERT fine-tuned on recruitment data, showing that fine-tuning sentence encoders on resume–vacancy matching improves recognition of professional skills, job requirements, and industry-specific terminology.

Despite their semantic strength, transformer-based models face real limitations in hiring contexts. Dense embedding models can assign high similarity scores to candidates whose resumes are contextually aligned with a job description while lacking essential technical qualifications. This has motivated hybrid architectures that combine semantic representations with explicit rule-based constraints, incorporating structured skill matching, keyword rules, or domain-specific validation

alongside neural similarity models. Combining lexical and semantic features allows a system to preserve important technical indicators, such as programming languages, certifications, and tools, while still recognizing semantically equivalent alternative phrasing.

Algorithmic bias is another major concern in automated recruitment. Tambe et al. [10] discuss how machine learning systems in human-resource management can reproduce historical biases when fairness is not deliberately incorporated during development and deployment. Raghavan et al. [11] further argue that simply removing demographic fields from applications is often insufficient, since models can infer sensitive characteristics through proxies such as educational background, employment history, linguistic patterns, or institutional affiliation. A common mitigation strategy is to preprocess resumes with Named Entity Recognition (NER) to identify and remove personally identifiable information before evaluation; the spaCy framework developed by Honnibal and Montani [6] provides widely used NER capabilities that support this form of anonymization.

The growing adoption of AI-driven recruitment has also increased the importance of model transparency. Complex neural architectures are frequently criticized as “black boxes,” making it difficult for recruiters and applicants to understand how a decision was reached. Ribeiro et al. [7] introduced Local Interpretable Model-Agnostic Explanations (LIME), which explains individual predictions by approximating model behavior through local perturbations. Ribeiro et al. [8] later extended model-agnostic interpretability with Anchors, which generate high-precision rule-based explanations that remain valid under local perturbation. Lundberg and Lee [12] proposed SHAP (SHapley Additive exPlanations), which applies cooperative game theory to quantify feature importance. While these post-hoc methods improve interpretability, they approximate model behavior rather than directly exposing the internal reasoning of the underlying model.

Information-retrieval-based recruitment systems are commonly evaluated with ranking metrics such as Normalized Discounted Cumulative Gain (NDCG), which measures ranking quality while emphasizing highly relevant candidates near the top of a ranked list. Järvelin and Kekäläinen [5] introduced NDCG as a robust evaluation framework for ranked retrieval, and it remains widely used for assessing recommendation and retrieval models.

Taken together, the literature highlights three persistent challenges in resume screening: achieving robust semantic understanding, mitigating algorithmic bias, and ensuring transparent decision-making. No single existing approach satisfies all three requirements simultaneously, which motivates hybrid recruitment frameworks that combine semantic embedding models with explicit rule-based verification to improve transparency, fairness, and accountability in automated hiring.

III. METHODOLOGY

A. System Overview

The proposed hybrid semantic-rule-based framework evaluates an unstructured candidate resume against a target job description through three sequential stages. The architecture is designed to capture semantic similarity through high-dimensional dense representations while simultaneously enforcing strict, rule-based verification of core technical competencies.

First, the resume undergoes identity-fair preprocessing to remove sensitive demographic information and reduce identity-related bias. Second, the cleaned text is analyzed through two parallel feature-extraction channels that evaluate lexical similarity and semantic similarity, respectively. Third, a rule-based skill-verification layer validates mandatory technical requirements and applies a fallback penalty when minimum competency thresholds are not met. The overall workflow is illustrated in Fig. 1.

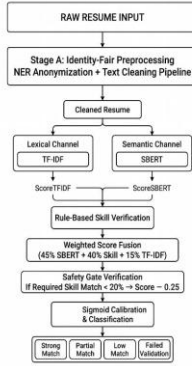


Fig. 1. Proposed system architecture.

Algorithm 1 summarizes the end-to-end ranking procedure.

Algorithm 1: Proposed Resume Ranking Framework

Input: Resume R , Job Description J
Output: Final Score, Candidate Category
Begin
1. Anonymize R using NER
2. Preprocess cleaned text
3. Compute lexical similarity score
4. Compute semantic similarity score
5. Compute S_{skill}
6. Compute $\text{Score}_{\text{RAW}}$
7. Apply Safety Gate
8. Compute $\text{Score}_{\text{Calibrated}}$
9. Assign Category
10. Return $\text{Score}_{\text{Calibrated}}$, Category
End

B. Preprocessing and Identity-Fair Anonymization

To minimize the influence of demographic information during evaluation, resumes are first processed through a Named Entity Recognition (NER) anonymization pipeline built on the spaCy framework [6]. Machine learning systems may unintentionally learn correlations tied to demographic attributes if identity markers remain present in the resume text. To reduce this risk,

the preprocessing pipeline automatically detects and suppresses named entities belonging to the categories PERSON, GPE (geopolitical entity), LOC (location), and DATE.

After entity detection, sensitive terms are replaced with standardized placeholder labels; the remaining text is lowercased, and stop words and unnecessary punctuation are removed. This preserves professional and technical content while reducing the influence of identity-related attributes. Table I illustrates the mapping between a raw input string and its identity-fair, cleaned counterpart.

TABLE I. RESUME PREPROCESSING AND IDENTITY-FAIR ANONYMIZATION MAPPING

Source Resume Text Segment (Raw Input)	Identity-Fair Scrubbed Output (Cleaned)	Removed Identity Entities
“John Doe, Senior Software Engineer based in New York. Completed contract on 05/12/2024...”	[PERSON], senior software engineer based in [GPE]. completed contract on [DATE]...	PERSON, GPE, DATE

The resulting cleaned text is used for all subsequent feature extraction and evaluation.

C. Dual-Channel Feature Extraction

Following preprocessing, the cleaned resume and job description are transformed into numerical representations through two independent channels.

1) Lexical Similarity Channel

The lexical channel measures direct vocabulary overlap using the TF-IDF representation proposed by Salton et al. [2]. Sublinear term-frequency scaling is applied during vectorization to reduce the influence of excessive keyword repetition. This channel captures exact terminology alignment between a resume and a job description and produces the lexical similarity score, denoted STF-IDF.

2) Semantic Similarity Channel

The semantic channel captures contextual similarity using Sentence-BERT (SBERT) [3]. The framework uses the all-MiniLM-L6-v2 bi-encoder to generate dense sentence embeddings; the resume and job description are independently encoded into fixed-dimensional vectors, and semantic similarity is computed via cosine similarity. This allows the framework to recognize equivalent skills and experience even when different terminology is used, producing the semantic similarity score, denoted SSBERT.

D. Rule-Based Skill Verification

Although semantic embeddings capture contextual meaning effectively, they may overestimate candidate suitability when strong semantic alignment is not backed by the required technical competencies. To address this, the framework incorporates a rule-based symbolic skill matrix. The job description is parsed to construct two skill dictionaries: Kreq,

representing mandatory technical skills, and K_{pref} , representing preferred skills.

Resume skills are matched against these dictionaries to compute the required-skill ratio (R_{match}) and preferred-skill ratio (P_{match}). The symbolic skill score is then computed as:

$$S_{skill} = (0.75 \times R_{match}) + (0.25 \times P_{match}) \quad (1)$$

This weighting prioritizes mandatory requirements while still accounting for additional desirable skills.

E. Score Fusion and Safety-Gate Mechanism

The lexical, semantic, and symbolic skill scores are combined into a single ranking value. Table II summarizes the composition and weighting of the score-fusion layer.

TABLE II. SCORE COMPOSITION AND WEIGHT DISTRIBUTION

Channel	Extraction Method	Optimization Role	Weight
Lexical (TF-IDF)	Sublinear TF-IDF	Exact keyword matching	15%
Semantic (SBERT)	SBERT (all-MiniLM-L6-v2)	Semantic context similarity	45%
Skill Matrix	Rule-based matching	Technical skill verification	40%

The final weighted fusion score is calculated as:

$$Score_{RAW} = 0.45(S_{SBERT}) + 0.40(S_{skill}) + 0.15(S_{TF-IDF}) \quad (2)$$

The weighting scheme was determined through iterative system tuning to balance contextual relevance against technical-skill verification.

To prevent technically underqualified candidates from achieving high rankings through semantic similarity alone, a safety-gate mechanism evaluates the required-skill fulfillment ratio (R_{match}). The final score is defined as:

$$Score_{Final} = \begin{cases} Score_{RAW} - 0.25, & \text{if } R_{match} < 0.20; \\ Score_{RAW}, & \text{otherwise} \end{cases} \quad (3)$$

This threshold acts as a minimum competency floor, lowering the ranking position of candidates who do not satisfy basic technical requirements. Unlike pure semantic ranking, the proposed framework explicitly penalizes candidates who fail to meet minimum mandatory-skill requirements, as illustrated conceptually in Fig. 2.

Figure 1: Pure Semantic Ranking with Required Skill Match Ratio

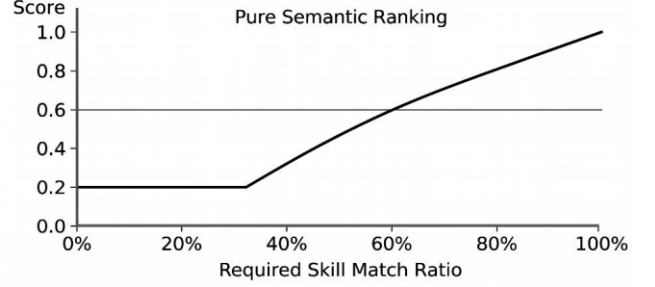


Fig. 2. Safety-gate penalty mechanism.

F. Score Calibration and Candidate Classification

The final score is normalized using a logistic sigmoid function:

$$Score_{Calibrated} = 1 / (1 + e^{-Score_{Final}}) \quad (4)$$

This transformation maps candidate scores to a bounded, probability-like range between 0 and 1. Candidates are then assigned to one of four ranking categories: Strong Match, Partial Match, Low Match, and Failed Validation. A candidate is classified as a Strong Match only when both the calibrated score and the prerequisite skill condition are satisfied:

$$StrongMatch = 1, \text{ if } Score_{Calibrated} \geq 0.65 \text{ and } R_{match} \geq 0.80; \text{ else } 0 \quad (5)$$

Candidates who exceed the calibrated score threshold but fail to satisfy the required-skill threshold are assigned to the Partial Match category, while profiles with lower calibrated scores are classified as Low Match or Failed Validation. The calibrated scores are used to rank candidates in descending order, enabling automated prioritization according to both semantic relevance and technical competency.

IV. EXPERIMENT AND ANALYSIS

A. Dataset and Evaluation Protocol

The base corpus for experimental evaluation was derived from a public resume dataset sourced from livecareer.com via Kaggle. While the complete dataset contains over 2,400 resumes across 24 professional categories, this study focuses on a subset of 120 Information Technology (IT) and software-engineering resumes. To evaluate ranking performance under realistic recruitment conditions, five independent enterprise job descriptions (JD_01–JD_05) were constructed for the roles of IT Support Specialist, Database Administrator, Web Developer, Systems Engineer, and Technical Support Engineer.

Evaluating the complete 120×5 candidate–job description grid would generate 600 resume–JD pairs, making exhaustive manual assessment susceptible to annotation fatigue and labeling inconsistency. A stratified sampling protocol was therefore adopted: for each job description, a preliminary ranking was generated and 15 resume–JD pairs were selected — the top 5 highest-ranked candidates, the bottom 5 lowest-ranked candidates, and 5 candidates sampled at random from

intermediate ranking positions — producing a manually labeled evaluation set of 75 distinct resume–JD pairs.

Each of the 75 sampled pairs was manually annotated by the authors using a three-level ordinal relevance scale: Label 2 (Strong Match) for candidates who demonstrate strong domain alignment and satisfy all essential technical requirements; Label 1 (Partial Match) for candidates with contextual relevance who lack one or more important technical competencies; and Label 0 (Low/No Match) for candidates with substantial domain misalignment or failure to satisfy critical technical requirements. Table III presents the distribution of the manually annotated evaluation dataset used for validation.

TABLE III. DISTRIBUTION OF MANUALLY LABELED RESUME–JOB DESCRIPTION EVALUATION SAMPLES

Target Job Description	Top (1–5)	Random	Bottom (71–75)	Total
JD_01: IT Support Specialist	5	5	5	15
JD_02: Database Administrator	5	5	5	15
JD_03: Web Developer	5	5	5	15
JD_04: Systems Engineer	5	5	5	15
JD_05: Technical Support Engineer	5	5	5	15
Total	25	25	25	75

B. Experimental Setup

All experiments were conducted locally using a Python-based implementation to preserve candidate privacy and support reproducible evaluation. Lexical features were generated with a TF-IDF vectorizer using sublinear logarithmic term-frequency scaling. Semantic representations were computed with the all-MiniLM-L6-v2 Sentence-BERT model from the sentence-transformers library, producing dense contextual embeddings without reliance on external cloud services.

The required-skill and preferred-skill dictionaries (Kreq and Kpref) were built from job-description requirements and converted into deterministic skill-verification rules. Three configurations were evaluated: (1) a Lexical-Only baseline using TF-IDF similarity with threshold $\tau = 0.12$; (2) a Semantic-Only baseline using SBERT similarity with threshold $\tau = 0.58$; and (3) the proposed Hybrid Framework, which integrates lexical similarity, semantic similarity, deterministic skill verification, and the safety-gate mechanism with its 20% required-skill threshold and penalty adjustment.

C. Performance Evaluation and Discussion

Ranking effectiveness was evaluated with standard information-retrieval metrics computed over the manually labeled 75-pair evaluation set. Mean NDCG measured ranking quality, Precision assessed shortlist purity, Recall measured

talent-discovery capability, and F1-score provided a balanced summary of classification performance.

TABLE IV. COMPARATIVE EVALUATION OF TF-IDF, SBERT, AND HYBRID RANKING MODELS

Metric	TF-IDF	SBERT	Hybrid
Mean NDCG	0.9582	0.9689	0.9366
Precision	1.0000	0.9762	0.9737
Recall	0.5500	0.6833	0.6167
F1-Score	0.7097	0.8039	0.7551

As shown in Table IV, the semantic-only SBERT baseline achieved the highest Mean NDCG (0.9689), reflecting the ability of dense contextual embeddings to capture semantic relationships beyond exact keyword matching. However, semantic similarity alone does not guarantee technical suitability: qualitative inspection revealed candidates with strong contextual alignment to a job description despite lacking critical technical competencies, underscoring a practical limitation of relying exclusively on semantic similarity for hiring decisions.

The comparative trends in Table IV are further illustrated in Fig. 3, which visually compares the TF-IDF baseline, SBERT baseline, and the proposed hybrid framework across all four evaluation metrics.

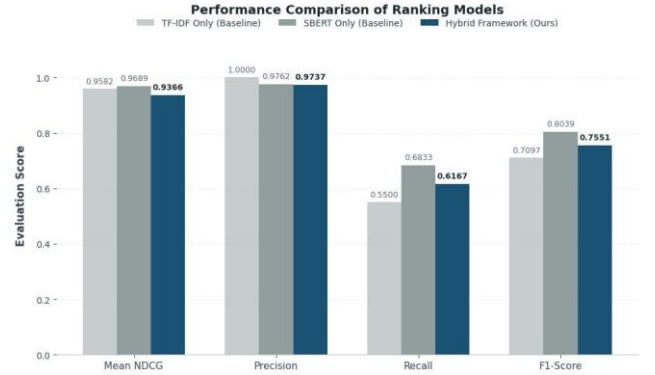


Fig. 3. Comparative performance of the TF-IDF baseline, SBERT baseline, and the proposed hybrid framework across Mean NDCG, Precision, Recall, and F1-score.

When the safety-gate mechanism and deterministic skill-verification layer were introduced, Mean NDCG decreased slightly to 0.9366. This reduction is an expected consequence of competency verification, which intentionally penalizes candidates who fail to satisfy minimum technical requirements; while it reduces ranking smoothness, it improves the framework’s ability to enforce qualification constraints relevant to real-world hiring.

A similar trade-off appears in the Precision–Recall relationship. The TF-IDF baseline achieved perfect Precision (1.0000) because it strongly favored exact lexical matches, but this strict behavior produced comparatively low Recall (0.5500), indicating that many potentially relevant candidates were

excluded due to lexical variation. The proposed hybrid framework improves Recall to 0.6167 while maintaining a high Precision of 0.9737, offering a more balanced trade-off between candidate retrieval and shortlist accuracy. These results suggest that combining semantic representations with explicit skill verification provides a practical balance between contextual relevance and competency validation in automated recruitment systems.

Although the evaluation dataset is relatively small, the stratified sampling strategy ensured representation across highly ranked, intermediate, and poorly ranked candidates, providing a balanced benchmark for comparative analysis while reducing annotation effort and evaluator fatigue. Future studies may further validate the proposed framework on larger and more diverse recruitment datasets.

V. CONCLUSION AND FUTURE WORK

Traditional Applicant Tracking Systems often struggle to balance semantic understanding with explicit technical-competency verification. Keyword-based approaches frequently reject qualified candidates because of lexical variation, whereas semantic embedding models may rank contextually similar resumes highly even when they lack essential job-specific skills [1], [3]. These limitations reduce the reliability and transparency of automated recruitment systems in practical hiring environments [10], [11].

This paper presented a hybrid resume-ranking framework that combines TF-IDF lexical matching, SBERT semantic embeddings, and deterministic rule-based skill verification to improve automated candidate screening. The architecture integrates semantic relevance with explicit competency validation through a weighted scoring mechanism, an identity-fair preprocessing layer, and a rule-based safety gate that penalizes candidates who fail to satisfy minimum technical-skill requirements, alongside transparent scoring and audit capabilities that improve the interpretability of automated recruitment decisions.

Experimental evaluation on a publicly available IT and software-engineering resume dataset, benchmarked against a manually annotated set of candidate-job description pairs, showed that while the semantic-only SBERT baseline achieved the highest Mean NDCG, the proposed hybrid framework provided a more balanced trade-off between semantic relevance and explicit technical-competency verification. These findings demonstrate that integrating semantic representations with deterministic skill verification offers a practical balance between ranking effectiveness, competency validation, and transparency in automated recruitment systems.

Although evaluation was performed on a relatively small manually annotated dataset, the stratified sampling protocol provided a balanced benchmark while reducing annotation effort. Future work will focus on validating the framework with larger and more diverse recruitment datasets, exploring adaptive weight-optimization techniques, and extending the system to support multilingual resume screening and cross-

domain recruitment scenarios. Additional explainability techniques, such as LIME [7] and SHAP [12], may also be incorporated to provide richer post-hoc explanations and further strengthen recruiter confidence in automated decision-making.

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